Question : 1

5-2 Draw the waveform of Q.

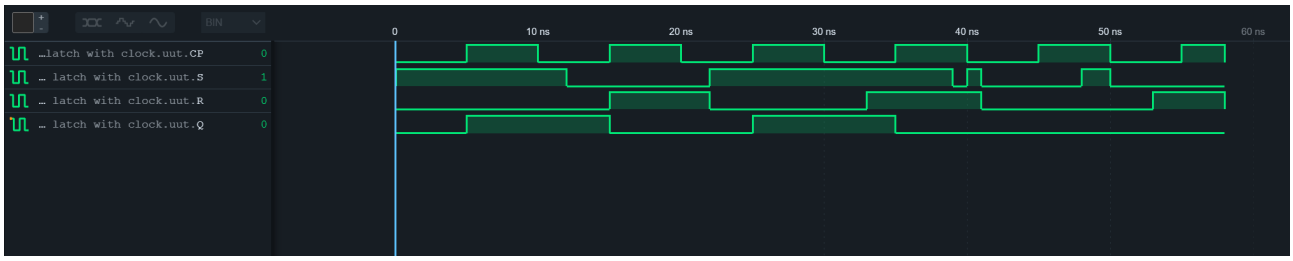
Answer:

由于 SR 锁存器的特征方程：

$$Q^{n+1} = S + \overline{R}Q^n$$

$$SR = 0$$

可知图形如下：



经过验证后输出正确。

Question : 2

5-3 Analyze the functions and the equation.

Answer:

(a) 由于 SR 锁存器的特征方程：

$$Q^{n+1} = S + \overline{R}Q^n$$

$$SR = 0$$

代入 $S = A\overline{Q}$ 与 $R = BQ$，得到：

$$Q^{n+1} = A\overline{Q^n} + \overline{BQ^n}Q^n$$

$$SR = 0$$

化简即为：

$$Q^{n+1} = A\overline{Q^n} + \overline{B}Q^n$$

$$SR = 0$$

(b) 同理，由 $S = C\overline{Q}$ 与 $R = D\overline{Q}$，得到：

$$Q^{n+1} = C\overline{Q^n} + \overline{D\overline{Q^n}}Q^n$$
$$SR = 0$$

化简即为:

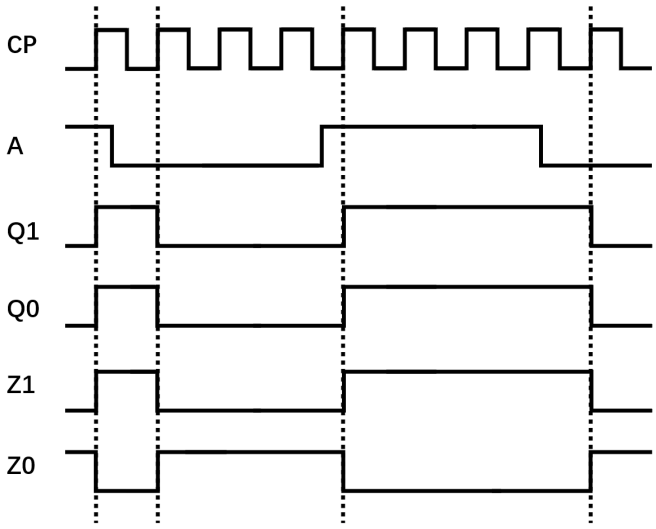
$$Q^{n+1} = C + Q^n$$
$$SR = 0$$

Question : 3

5-9 Draw the waveform of Q and Z.

Answer:

由 D 触发器的公式 $Q^{n+1} = D$ 得到如图所示:



Question : 4

5-10 Draw the waveform of Q and Z.

Answer:

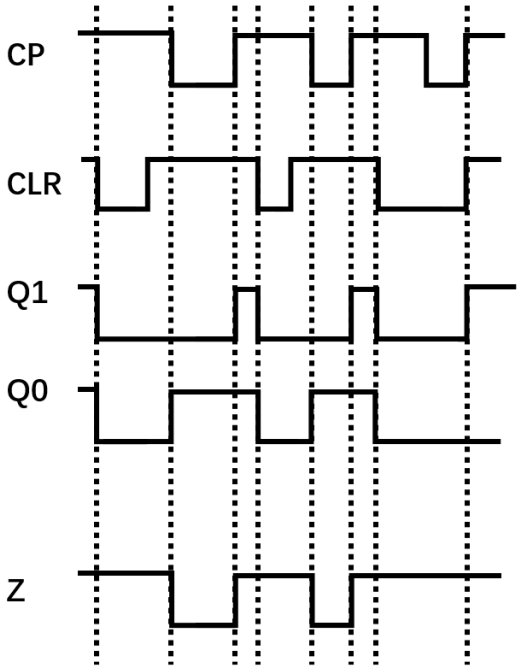
由 JK 触发器的公式 $Q^{n+1} = J\overline{Q^n} + \overline{K}Q^n$，化简得到:

$$Q_0^{n+1} = \overline{Q_0^n} + Q_0^n = 1$$

$$Q_1^{n+1} = \overline{Q_1^n} + Q_1^n = 1$$

$$Z = \overline{Q_0^{n+1}\overline{Q_1^{n+1}}} = \overline{Q_0^{n+1}} + Q_1^{n+1}$$

得到如图所示：

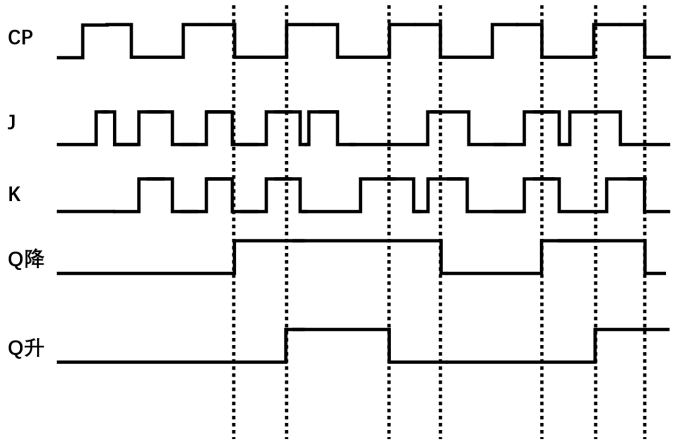


Question : 5

5-13 Draw the waveform of Q.

Answer:

由 JK 触发器的公式 $Q^{n+1} = J\overline{Q^n} + \overline{K}Q^n$ 得到如图所示：

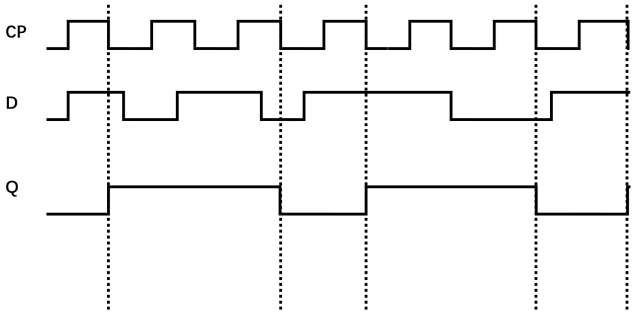


Question : 6

5-14 Draw the waveform of Q.

Answer:

同理，根据题目所述，绘制图如下：

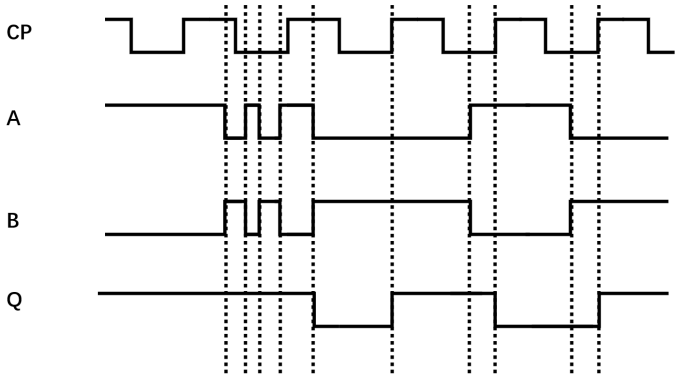


Question : 7

5-18 Draw the waveform of B and Q.

Answer:

同理可得其状态转移方程，根据题目所述，绘制如图：



Question : 8

5-21 Draw the waveform and find the mistake of the code.

Answer:

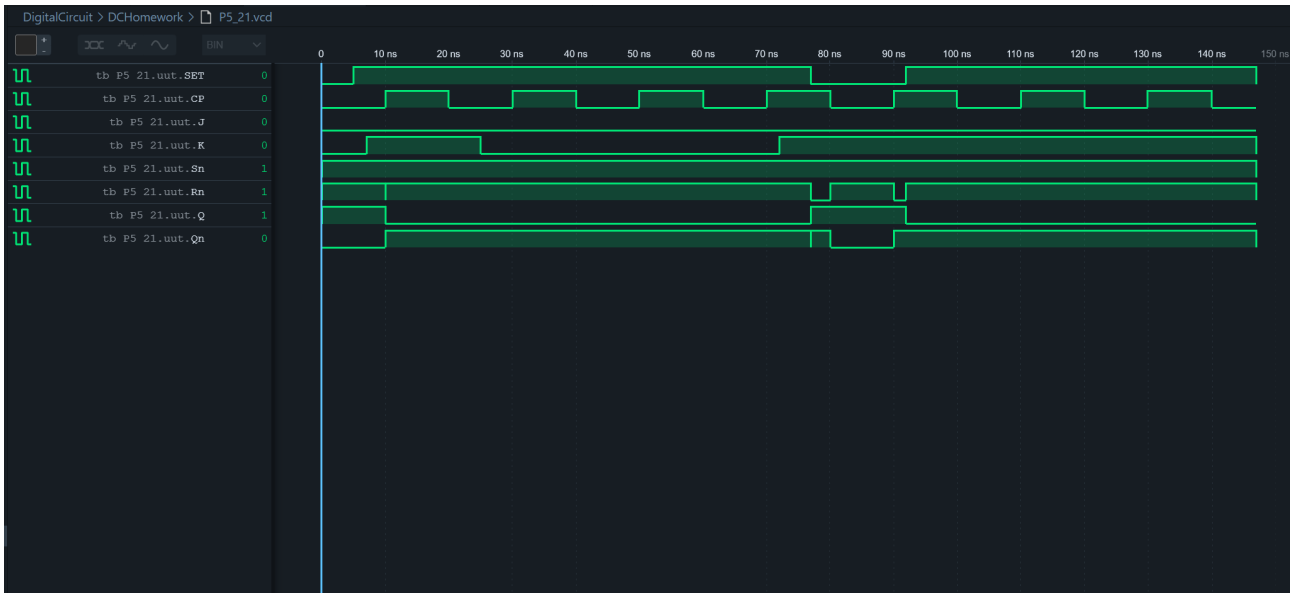
问题在于，当 $J = 1$ 和 $K = 1$ 时：假设 CP 为 1：

$$WRn = (1\&1\&WQ) = WQ$$
$$WSn = (WQn\&1\&1) = WQn$$

反馈回路的竞争可能会导致 WQ 和 WQn 之间不断地翻转，这会使仿真工具进入无穷循环，导致 VCD 文件卡住。

解决方案：在某些情况下，通过添加延迟或状态检测，可以避免竞争状态或震荡。可以修改测试平台，以便在特定状态下暂停或确保在切换之前有足够的时间进行稳定，或者修改模块，在 J=1 和 K=1 的情况下，增加适当的逻辑以防止输出的不稳定。或者，使用寄存器来存储并输出稳定状态，根据实际需求调整逻辑处理，以确保模块能够正常工作。

忽略 $J = 1$ 和 $K = 1$ 时的波形图如下：

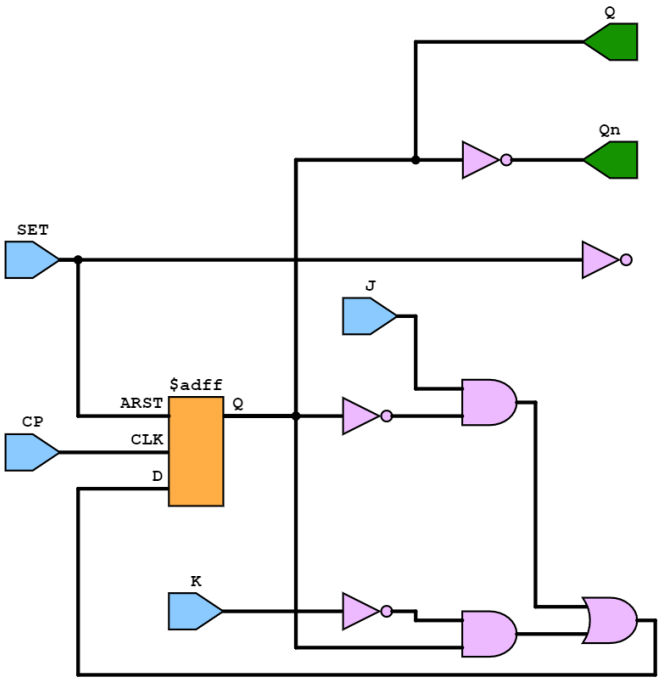


Question : 9

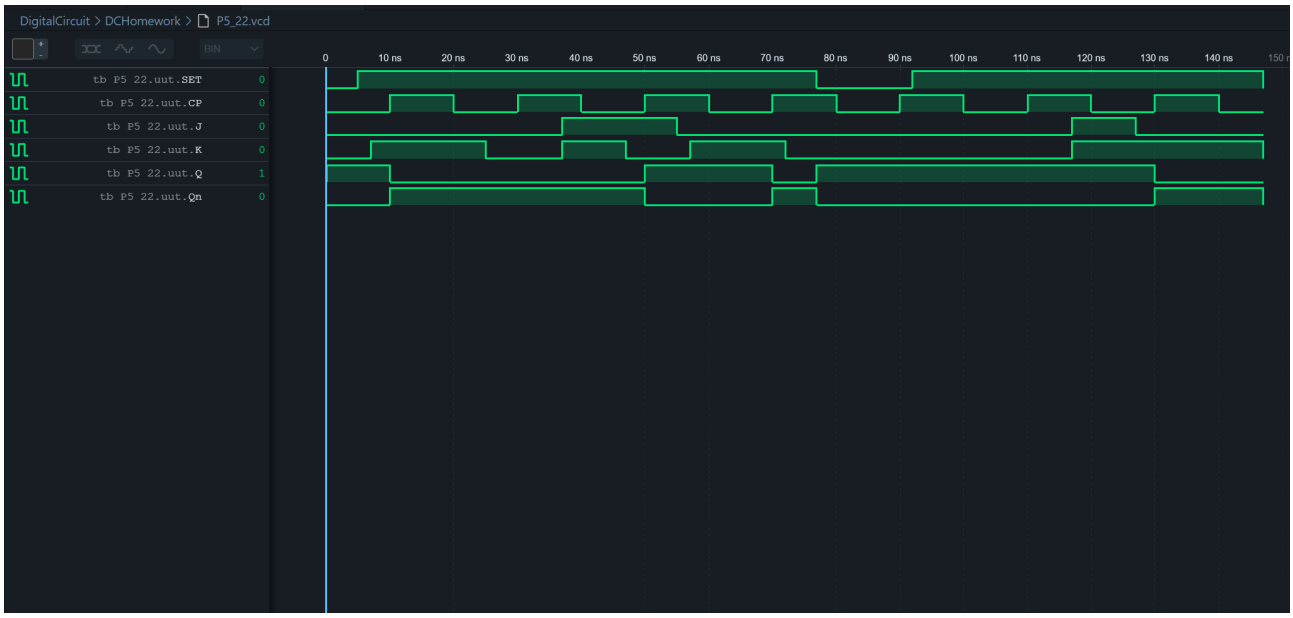
5-22 Draw logic diagram and waveform of the module.

Answer:

由题目易得，逻辑图如下：



波形图如下：



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感谢您的批阅!